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An Analysis of Economic Load Dispatch with Ramp-rate limit constraints using Different Algorithms

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Abstract*— The economic load dispatch (ELD) plays an important role in power system operation and control. Different techniques have been used to solve these problems. Recently, the soft computing techniques have widely used in practical applications. This paper shows successful implementation of four evolutionary algorithms, namely particle swarm optimization (PSO), particle swarm optimization with constriction factor approach (PSOCFA), particle swarm optimization with inertia weight factor approach (PSOIWA) and particle swarm optimization with constriction factor and inertia weight factor approach (PSOCFIWA) algorithms. Here ramp-rate limit constraints are considered to solve this problem. Here three cases have been considered. Four algorithms have been applied for each case. Optimum fuel cost is taken from each algorithm is compared for each cases (*Case study I, II and III).

Keywords*— Economic Load Dispatch, PSO, PSOCFA, PSOIWA, PSOCFIWA, Ramp-rate limit.

I. INTRODUCTION

Economic load dispatch (ELD) problem is the one of the most important optimization problem in power system. Economic load dispatch in power system is a task of allocating power generation among the committed thermal units and satisfied the load demand such as that the cost of production is minimized and constrained are satisfied at the same time. Here Ramp-rate limit constraints are considered to solve this problem.

Traditionally in ELD problems, the cost function for generating units has been approximated as a quadratic function. Practically these cost functions of modern power generation units are highly nonlinear, and discrete in nature.

A bibliographical survey on ELD methods reveals that various numerical optimization techniques have been employed to solve the ELD problem. ELD is solved traditionally using mathematical programming based on optimization techniques such as Genetic Algorithm [1-3] , Differential Evolution [4-8] Evolutionary programming [9-12], Simulated annealing [13-15], Artificial neural networks [16-19], and Taboo search [20-22].

ELD problems have been recently solved by Particle Swarm Optimization (PSO) approaches [23-27]. The PSO originally developed by Eberhart and Kennedy [23-25] in 1995, which is a population based stochastic algorithm.

Literature survey shows that particle swarm optimization technique is very simple optimization technique and easier to understand from any other techniques. So this algorithm has drawn a wide attention in the world. This optimization technique has a faster convergence rate to global solution. It has highest potential of finding more nearly global solutions to the Economic load Dispatch Problems [27, 28].

PSO has small number of free tunable parameters to arrive desire goal. PSO has been applied in many field of power system to determine the optimum value of the parameter of power system.

This paper proposes particle swarm optimization (PSO), particle swarm optimization with constriction factor approach (PSOCFA), particle swarm optimization with inertia weight factor approach (PSOIWA) and particle swarm optimization with constriction factor and inertia weight factor approach (PSOCFIWA) to solve ELD problem with three, six and fifteen thermal units with generator constraints. The application of PSO, PSOCFA, PSOIWA and PSOCFIWA algorithms to three, six and fifteen-generator ELD problems has been successfully demonstrated.

II. PROBLEM FORMULATION

The objective of the economic dispatch problem is to minimize the total fuel cost while satisfying the constraints.

A. Objective function

Fuel cost function of each thermal generating unit is expressed as a quadratic function. In terms of real power output, total cost can be expressed as the following

$$\min f = \sum_{i=1}^{n} a_i P_i^2 + b_i P_i + c_i \quad i = 1,2,3.....,n \tag{1}$$

Where n is the number of generating units, is the real power output (MW) of generator and, are cost coefficients.

B. Constraints

(i) Power balance constraints

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Generated power should be the same as the total load demand P_D (in MW). In this case, the active power balance is given by

$$P_D = \sum_{i=1}^{n} P_i \tag{2}$$

The transmission and generator losses have been neglected.

(ii) Generation limits

The output power of each generating unit has a lower and upper bound so that it lies in between these bounds.

$$P_i^{\min} \le P_i \le P_i^{\max} \tag{3}$$

Where $P_i^{\min}$ and $P_i^{\max}$ are the minimum and maximum output of generator i, respectively.

(iii) Ramp-rate limit constraints

$$\max\left(P_i^{\min}, P_i^0 - DR_i\right) \le P_i \le \min\left(P_i^{\max}, P_i^0 + UR_i\right) \tag{4}$$

Where $P_i(t)$ is the present output power, P_i^0 is the previous output power, UR_i is the up-ramp limit of the ith generator (in units of MW/time-period), and DR_i is the down-ramp limit of the ith generator (in units of MW/time-period).

III. Particle Swarm Optimization (PSO)

PSO, first developed by Eberhart and Kennedy (1995), is a population based optimization algorithm. The population is called 'swarm'. Each potential solution is called particle which is given a random velocity and is flown through the solution space (similar to the search process for food of a bird swarm) searching for the optimal position. Each particle keeps track of its previous best position, called *pbest*, and corresponding fitness in its memory. The best value of pbest is called gbest, which is the best position discovered by the swarm. If promising new solution is discovered by a particle then all other particles will move closer to it. Based on PSO concept, mathematical equations for the searching process are:

A. Velocity updating equation:

$$v_{id}^{(k+1)} = w \times v_{id}^{(k)} + C_1 \times rand(\ ) \times \left(pbest_{id} - x_{id}^k\right) + C_2 \times rand(\ ) \times \left(gbest_{id} - x_{id}^k\right) \tag{5}$$

B. Position updating equation:

$$x_{id}^{(k+1)} = x_{id}^k + v_{id}^{(k+1)} \tag{6}$$

where x_{id}^k, x_{id}^{k+1} are the position of dth dimension (variable) of the ith particle at k^{th} and $(k+1)^{th}$ iteration, v_{id}^k, v_{id}^{k+1} are the velocity of the dth dimension of the ith particle at the kth and the $(k+1)^{th}$ iteration and C_1, C_2 are the cognitive and the social parameters; $r1$ and $r2$ are random numbers uniformly distributed within [0, 1]; $pbest_{id}$ is the best position of the dth dimension of the ith particle; $gbest_d$ is the group best position of the dth dimension and w is the inertia weight factor.

$$w = w_{\max} - \frac{w_{\max} - w_{\min}}{iter_{\max}} \times iter \tag{7}$$

Where $iter_{\max}$ is the maximum number of iterations and $iter$ is the current number of iterations.

IV. Particle Swarm Optimization With Constriction Factor Approach (PSOCFA)

For particle swarm optimization with constriction factor approach (PSOCFA), the velocity of equation no. (6) is manipulated in accordance with equation no. (9).

$$v_{id}^{(k+1)} = CFa \times \begin{pmatrix} w \times v_{id}^{(k)} + C_1 \times rand(\ ) \times \left(pbest_{id} - x_{id}^k\right) \\ + C_2 \times rand(\ ) \times \left(gbest_{id} - x_{id}^k\right) \end{pmatrix} \tag{8}$$

Constriction factor (CFa) varies from 0.60 to 0.73.

V. Particle Swarm Optimization With Inertia Weight Factor Approach (PSOIWA)

For particle swarm optimization with inertia weight factor approach (PSOIWA), the velocity of equation no. (5) is manipulated in accordance with equation no. (10). In inertia weight factor approach (IWA), inertia weight (w^{k+1}) at $(k+1)^{th}$ cycle is given by

$$w^{k+1} = \frac{w_{\max} - \left(w_{\max} - w_{\min}\right)}{k_{\max}} \times (k+1) \tag{9}$$

Velocity updating equation:

$$v_{id}^{(k+1)} = w^{k+1} \times v_{id}^{(k)} + C_1 \times rand(\ ) \times \left(pbest_{id} - x_{id}^k\right) + C_2 \times rand(\ ) \times \left(gbest_{id} - x_{id}^k\right) \tag{10}$$

Where $w_{\max}$ =1; $w_{\min}$ =0.4; $k_{\max}$ = maximum no of iteration cycle.

VI. Particle Swarm Optimization With Constriction Factor And Inertia Weight Factor Approach (PSOCFIWA)

For particle swarm optimization with constriction factor and inertia weight factor approach (PSOCFIWA), the velocity of equation no. (6) is manipulated in accordance with the following equation

$$v_{id}^{(k+1)} = CFa \times \begin{pmatrix} w^{k+1} \times v_{id}^{(k)} + C_1 \times rand(\ ) \times \left(pbest_{id} - x_{id}^k\right) \\ + C_2 \times rand(\ ) \times \left(gbest_{id} - x_{id}^k\right) \end{pmatrix} \tag{11}$$

Here also, Constriction factor (CFa) varies 0.6 to 0.73 and the inertia weight factor approach (IWA) follows equation (9).

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VII. Implementation Of PSO, PSOCFA, PSOIWA And PSOCFIWA Methods In ELD Problem

This paper presents a solution of economic dispatch problem using PSO, PSOCFA, PSOIWA and PSOCFIWA algorithm. Its implementation consists of the following steps:

Step 1 The particles are randomly generated between the operating limits.

Step 2 The values of the fitness function of the particles are evaluated using objective function (equation 1) and the dimensions (variables) of the particles are initialized as *pbest (s)*.

Step 3 The best value of *pbest(s)* is represented as *gbest*.

Step 4 The particles' velocities and positions are updated using velocity and position updating equations [equation nos. (5) and (6) for PSO technique, equation nos. (8) and (6) for PSOCFA technique, equation nos. (10) and (6) for PSOIWA technique and equation nos. (11) and (6) for PSOCFIWA technique].

Step 5 The new fitness function values are evaluated using the updated positions of the particles. If the current position of the particle is better than its previous *pbest*, the *pbest* is updated by the current particle, otherwise not updated. The updated *gbest* is the best among all the *pbest(s)*.

Step 6 If the stopping criterion is satisfied, go to Step 7, otherwise, go to Step 2.

Step 7 The particle that generates the latest *gbest* yields the optimal variables.

VIII. Simulation Results

The economic load dispatch (ELD) problem [27, 28] has been solved using the PSO, PSOCFA, PSOIWA and PSOCFIWA algorithms. To assess the efficiency of the proposed approaches, three case studies of ELD problems have been considered. Valve point loading and losses has not been considered in objective functions for simplicity. Ramp-rate limit constraints has been considered in ELD problem. In simulation, population size=100, maximum iteration = 500 has been considered. Table I, table II and table III show data for the three, six and fifteen thermal units of generating unit capacity, ramp-rate limits and cost coefficients. Table IV, Table V and Table VI show the comparing results of PSO, PSOCFA, PSOIWA and PSOCFIWA for case study I, II and III.

A. Case study I: Three-unit system

This case study consists of three thermal units [27]. All thermal units are within minimum and maximum generating limits. Generating unit capacity and coefficients of three thermal units are given in Tables I. In this case, the load demand is P_D = 850 MW.

Table I: Data for the three thermal units of generating unit capacity and coefficients

Unit	P_i^{min}	P_i^{max}	$p_i^{\bullet}$	UR_i	DR_i	a_i	b_i	c_i
1	50	200	170	50	90	0.004820	7.97	78
2	100	400	350	80	120	0.001940	7.85	310
3	100	600	440	80	120	0.001562	7.92	562

B. Case study II: Six-unit system

This case study consists of six thermal units [27]. All thermal units are within minimum and maximum generating limits. Generating unit capacity and coefficients of six thermal units are given in Tables II. In this case, the load demand is P_D = 1263 MW.

Table II: Data for the six thermal units of generating unit capacity and coefficients

Unit	P_i^{min}	P_i^{max}	$p_i^{\bullet}$	UR_i	DR_i	a_i	b_i	c_i
1	100	500	440	80	120	0.0070	7	240
2	50	200	170	50	90	0.0095	10	200
3	80	300	200	65	100	0.0090	8.5	220
4	50	150	150	50	90	0.0090	11	200
5	50	200	190	50	90	0.0080	10.5	220
6	50	120	110	50	90	0.0075	12.0	190

C. Case study III: Fifteen-unit system

This case study consists of fifteen thermal units [28]. All thermal units are within minimum and maximum generating limits. Generating unit capacity and coefficients of fifteen thermal units are given in Tables III. In this case, the load demand is P_D = 2630 MW.

Table III: Data for the fifteen thermal units of generating unit capacity and coefficients

Unit	P_i^{min}	P_i^{max}	$p_i^{\bullet}$	UR_i	DR_i	a_i	b_i	c_i
1	150	455	400	80	120	0.0003	10.1000	671
2	150	455	300	80	120	0.0002	10.2000	574
3	20	130	105	130	130	0.0011	8.8000	374
4	20	130	100	130	130	0.0011	8.8000	374
5	150	470	90	80	120	0.0002	10.4000	461
6	135	460	400	80	120	0.0003	10.1000	630
7	135	465	350	80	120	0.0004	9.8000	548
8	60	300	95	65	100	0.0003	11.2000	227
9	25	162	105	60	100	0.0008	11.2000	173
10	25	160	110	60	100	0.0012	10.7000	175
11	20	80	60	80	80	0.0036	10.2000	186
12	20	80	40	80	80	0.0055	9.9000	230
13	25	85	30	80	80	0.0004	13.1000	225
14	15	55	20	55	55	0.0019	12.1000	309
15	15	55	20	55	55	0.0044	12.4000	323

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Table IV: Comparison of four methods: best result for the case study I

Unit of power output	PSO	PSOCFA	PSOIWA	PSOCFIWA
P_1	122.2264	122.2264	122.2264	122.2264
P_2	334.6037	334.6038	334.6038	334.6038
P_3	393.1698	393.1698	393.1698	393.1698
Total cost($/hr)	8195.4	8195.4	8195.4	8195.4
CPU Time(sec.)	0.453800	0.458900	**0.420686**	0.459870

Table V: Comparison of four methods: best result for the case study II

Unit of power output	PSO	PSOCFA	PSOIWA	PSOCFIWA
P_1	446.7073	446.5004	446.7073	446.6612
P_2	171.2580	171.1056	171.2580	171.6678
P_3	264.1057	265.0000	264.1057	263.7700
P_4	125.2168	125.0559	125.2168	125.4466
P_5	172.1189	171.9378	172.1189	171.5853
P_6	83.5934	83.4004	83.5935	83.8692
Total cost($/hr)	15,276	15,276	15,276	15,276
CPU Time(sec.)	0.560481	0.545833	**0.534183**	0.548816

Table VI: Comparison of four methods: best result for the case study III

Unit of power output	PSO	PSOCFA	PSOIWA	PSOCFIWA
P_1	455.0000	454.9999	455.0000	432.2206
P_2	380.0000	380.0000	380.0000	380.0000
P_3	130.0000	130.0000	129.9978	129.9956
P_4	129.9997	129.9999	130.0000	125.4094
P_5	170.0000	170.0000	170.0000	160.3348
P_6	460.0000	460.0000	460.0000	451.4987
P_7	430.0000	430.0000	430.0000	430.0000
P_8	60.0000	64.0539	75.0022	98.4596
P_9	40.0001	35.9463	25.0000	116.1367
P_{10}	160.0000	160.0000	160.0000	124.0749
P_{11}	80.0000	80.0000	80.0000	53.4100
P_{12}	80.0000	80.0000	80.0000	71.1366
P_{13}	25.0000	25.0000	25.0000	26.5963
P_{14}	15.0002	15.0000	15.0000	15.7267
P_{15}	15.0000	15.0001	15.0000	15.0000
Total cost($/hr)	32,359	32,359	32,359	32,441
CPU Time(sec.)	5.931256	6.158593	**5.547443**	5.941222

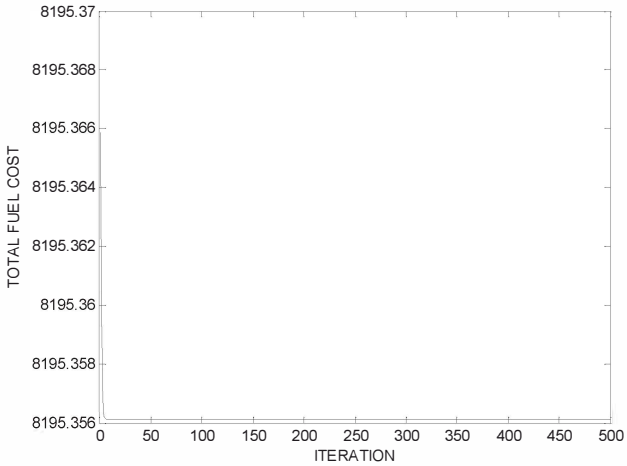

Fig. 1 Convergence Behavior of the PSOIWA Method for 3-Unit System (PD=850MW)

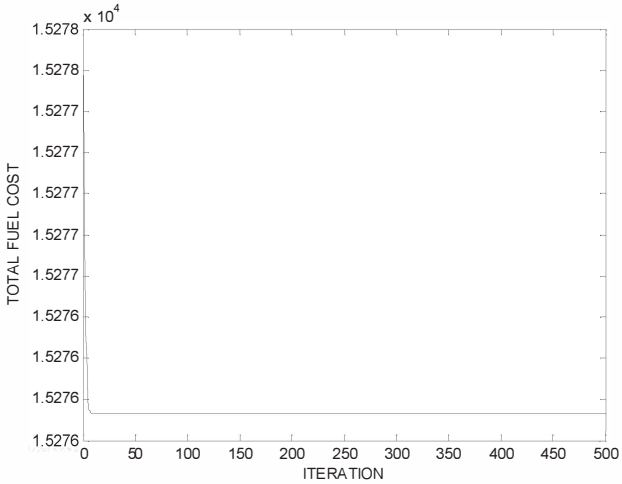

Fig. 2 Convergence Behavior of the PSOIWA Method for 6-Unit System (PD=1263MW)

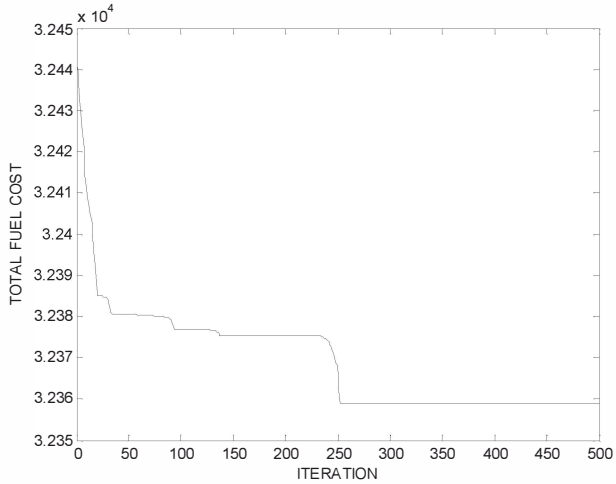

Fig. 3 Convergence Behavior of the PSOIWA Method for 15-Unit System (PD=2630MW)

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IX. CONCLUSION

In this paper PSO, PSOCFA, PSOIWA and PSOCFIWA are applied to solve ELD problems. The application of PSO, PSOCFA, PSOIWA and PSOCFIWA algorithms to three, six and fifteen generator ELD problem has been successfully demonstrated. Here, ramp-rate limits constrain has been considered in economic load dispatch problem. From the results it is clear that PSOIWA approach gives the best global optimum solution with less computation time than the PSO, PSOCFA and PSOCFIWA techniques. The test results show that PSOIWA algorithm converge to optimal fuel cost.

REFERENCES

[1] Tofighi, M. ; Maddahi, R. ; Sadeqzadeh, M. An improved genetic algorithm based economic dispatch with nonsmooth fuel cost function, ICEEI.2011, pp. 1-6, 2011.

[2] Ling, S.H. ; Lam, H.K. ; Leung, F.H.F. ; Lee, Y.S. , Improved genetic algorithm for economic loaddispatch with valve-point loadings, Industrial Electronics Society, 2003. IECON '03. pp. 442 - 447 , 2003.

[3] Chao-Lung Chiang, Improved genetic algorithm for power economic dispatch of units with valve-point effects and multiple fuels, Power Systems, IEEE Transactions on Volume: 20 , Issue: 4, pp.1690-1699, 2005.

[4] Lai, J.C. ; Leung, F.H. ; Sai-Ho Ling , Economic Load Dispatch using Differential Evolution with double wavelet mutation operations, Evolutionary Computation (CEC), 2010 IEEE, pp. 1-6, 2010.

[5] Das, A.K. ; Majumdar, R. ; Krishnanand, K.R. ; Panigrahi, B.K. Economic load dispatch using hybridized Differential Evolution and invasive weed operation, Energy, Automation, and Signal (ICEAS), 2011, pp. 1 – 5, 2011.

[6] Sharma, R. ; Samantaray, P. ; Mohanty, D.P. ; Rout, P.K., Environmental economic load dispatch using multiobjective differential evolution algorithm,ICES 2011, pp. 1-7, 2011.

[7] Sharma, R. ; Panigrahi, K.B. ; Rout, P.K. ; Krishnanand, K.R., A solution to economic load dispatch problem with non-smooth cost function using Self-Realized Differential Evolution optimization algorithm, ICEAS.2011, pp. 1-6, 2011.

[8] Saber, A.Y. ; Rahman, D.M.F., Economic load dispatch using particle swarm differential evolution optimization, Power and Energy Society General Meeting, PES.2011 IEEE, pp. 1-8, 2011.

[9] Suganya, G. ; Balamurugan, K. ; Dharmalingam, V. Multi-objective evolutionary programming technique for economic/emission load dispatch, ICAESM 2012,IEEE, pp. 269- 275, 2012.

[10] Jayabarathi T, Sadasivam G, Ramachandran V. Evolutionary programming based economic dispatch of generators with prohibited operating zones. Elect Power Syst Res 52(3):pp. 261–6 , 1999.

[11] Sinha N, Chakrabarti R, Chattopadhyay PK. Evolutionary programming techniques for economic load dispatch. IEEE Trans Evol Comput;7(1):pp. 83–94, 2003.

[12] Pereira-Neto A, Unsihuay C, Saavedra OR. Efficient evolutionary strategy optimisation procedure to solve the nonconvex economic dispatch problem with generator constraints. IEEE Proc Gener Transm Distrib;pp. 653–60, 152(5), 2005.

[13] Vanitha, M. ; Thanushkodi, K. , Solving non-convex economic load dispatch problem by Efficient Hybrid Simulated Annealing algorithm, Advanced Communication Control and Computing Technologies (ICACCCT), 2012 IEEE International Conference, pp. 362-365, 2012.

[14] Wong KP, Wong YW. Genetic and genetic/simulated-annealing approaches to economic dispatch. IEEE Proc Gener Transm Distrib 1994; pp. 507–13, 141(5).

[15] Roa-Sepulveda CA, Pavez-Lazo BJ. A solution to the optimal power flow using simulated annealing. Electr Power Energy Syst;pp. 25(1):47–57, 2003.

[16] Kumarappan, N. ; Mohan, M.R. ; Murugappan, S , ANN approach applied to combined economic and emission dispatch for large-scale system, IJCNN '02., pp 323-327, 2002.

[17] Panta, S. ; Premrudeepreechacharn, S., Economic dispatch for power generation using artificial neural network ICPE'07 conference in Daegu, Korea, pp. 558-562, 2007.

[18] Kumar J, Seble´ GB. Clamped state solution of artificial neural network for real-time economic dispatch. IEEE Trans Power Syst;pp. 10(2):925–31, 1995.

[19] Yalcinoz T, Short MJ. Neural network approach for solving economic dispatch problem with transmission capacity constraints. IEEE Trans Power Syst ;pp. 13(2):307–13, 1998.

[20] Lin WM, Chen FS, Tsay MT. An improved tabu search for economic dispatch with multiple minima. IEEE Trans Power Syst 2002;17(1):108–112.

[21] Tippayachai, J. ; Ongsakul, W. ; Ngamroo, I. , Nonconvex economic dispatch by enhanced tabu search algorithm, Power Engineering Society General Meeting, 2003, IEEE, PES.2003, Volume: 2, 2003.

[22] Khamsawang S, Boonseng C, Pothiya S. Solving the economic dispatch problem with tabu search algorithm. In: IEEE International Conference on Industrial Technology, Bangkok, Thailand. pp. 274–278, 2002.

[23] Khamsawang, S. ; Wannakarn, P. ; Jiriwibhakorn, S. , Hybrid PSO-DE for solving the economic dispatch problem with generator constraints, Computer and Automation Engineering (ICCAE), 2010, pp. 135-139. 2010.

[24] Hamed, M. ; Mahdad, B. ; Srairi, K. ; Mancer, N. , Solving combined economic emission dispatch problem using time varying acceleration based PSO, Environment and Electrical Engineering (EEEIC), 2012, pp. 764- 769, 2012.

[25] Varma, S.C. ; Murthy, K.S.L. ; SriChandan, K. , Gaussian Particle swarm optimization for combined economic emission dispatch, Energy Efficient Technologies for Sustainability (ICEETS), 2013, pp. 1336-1340, 2013.

[26] Rani, C. ; Kothari, D.P. ; Busawon, K.Combined economic emission dispatch problem using chaotic self adaptive PSO, Power, Energy and Control (ICPEC), 2013 International Conference, pp. 399- 404, 2013.

[27] Singh Nagendra, Kumar Yogendra, Modarate Random PSO using for Economic Load Dispatch, IJERT, Vol.1, Issue 7, September 2012.

[28] Leandro dos Santos Coelho , Chu-Sheng Lee Solving economic load dispatch problems in power systems using chaotic and Gaussian particle swarm optimization approaches, In: Electrical Power and Energy Systems 30 pp. 297–307, 2008.

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